General Mathematics — Exercise Sheet 36

Differentiability in several variables

Variant: Statements

Exercise 1 (A radial power and the full remainder). For each $p \in \mathbb{R}$, define $f_p(0,0) = 0$ and $f_p(x,y) = (x^2+y^2)^{p/2}$ when $(x,y) \neq (0,0)$. Determine exactly when f_p is continuous at the origin, when both its coordinate partial derivatives exist there, when it is differentiable there, and when it is C^1 on $\mathbb{R}^2$. Use Lecture 36's full remainder definition and Lecture 8's real-power derivative. Do not infer differentiability from coordinate quotients alone.

Exercise 2 (Differentiability without continuous partial derivatives). Set $f(0,y) = 0$ for every $y \in \mathbb{R}$ and $f(x,y) = x^2 \sin(1/x)$ for $x \neq 0$. Prove differentiability at the origin directly and find its derivative. Compute both partial derivatives near the origin, including on the line $x = 0$, and show that $\partial_x f$ is not continuous at the origin. Explain why this does not contradict Lecture 36's continuity criterion. Use Lecture 8's differentiation rules and Lectures 3 and 5's sequence and function limits.

Exercise 3 (Two nonlinear maps and the matrix order). Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ and $G : \mathbb{R}^3 \to \mathbb{R}^2$ be given by

$$F(x,y) = (e^{xy}, x + y^2, x^2 - y), \qquad G(\xi,\eta,\zeta) = (\xi\eta, \eta + \zeta^2).$$

Find $F(0,1)$, $J_F(0,1)$, $J_G(F(0,1))$ and $J_{G\circ F}(0,1)$. Justify differentiability, and verify the final matrix by differentiating the directly substituted composition at that point. Use Lecture 36's continuity criterion and matrix chain rule, and Lecture 19's matrix product.

Exercise 4 (Directional changes under a direction constraint). For $f(x,y,z) = xy + yz + zx$ at $P = (1,2,-1)$, consider unit directions $u = (u_1,u_2,u_3)$ subject to $u_2 = c$, where $c \in \mathbb{R}$ is prescribed. Determine when such directions exist. Whenever they do, find the maximum and minimum of $D_u f(P)$ and all directions attaining them. Include the boundary values $c = -1$ and $c = 1$. Use Lecture 36's directional formula and Lecture 12's Cauchy–Schwarz inequality.

Exercise 5 (A level curve without an implicit existence theorem). Let $f(x,y) = x^2y + y^3$ and $P = (1,1)$. Suppose a differentiable regular curve through P lies in the level set $f = 2$. Find its possible tangent line at P by Lecture 36's chain rule. Then construct an actual regular curve through P on that level by taking $y = t$ and the positive square root for x, on an open interval about $t = 1$. Verify its domain, equation and tangent directly. Use Lecture 8's square-root derivative and Lecture 36's gradient-normal result. Do not invoke Lecture 37's implicit function theorem.

Exercise 6 (An orthogonal coordinate change ⋆). Let Q be a real $n \times n$ matrix satisfying $Q^{\mathsf{T}}Q = I$. For a scalar function f defined near $a \in \mathbb{R}^n$, set $g(h) = f(a + Qh)$ for h near zero. Prove that f is differentiable at a exactly when g is differentiable at zero. Prove the gradient transformation $\nabla g(0) = Q^{\mathsf{T}}\nabla f(a)$ and equality of the gradient norms. Apply this to $f(x,y) = e^{x+2y}$, $a = (0,0)$ and $Q = \frac{1}{5}\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$. Use Lectures 19 and 21's matrix algebra/invertibility, Lecture 12's norm, and Lecture 36's chain rule. The matrix condition defines an orthogonal change of coordinates.